Practical no:- 30
SCALA for DS

Name :-Anshu Sharma	Roll no:-35
Class:- SYIT	Subject :-SCALA for DS
Date of assignment:-29-09-26	Date of submission:-29-09-26

Aim: Cluster a dataset into two groups using k-means clustering in Breeze.
Input:

```
import breeze.linalg._
object Demo5 extends App {  new *
  // Dataset
  val data = DenseVector(1.0, 2.0, 3.0, 10.0, 11.0, 12.0)

  // Initial cluster centers
  var center1 = 2.0
  var center2 = 10.0

  // Repeat clustering
  for (i <- 1 ≤ to ≤ 5) {
    val cluster1 = data.toArray.filter(x =>
      math.abs(x - center1) < math.abs(x - center2)
    )

    val cluster2 = data.toArray.filter(x =>
      math.abs(x - center2) <= math.abs(x - center1)
    )
    // Calculate new centers
    center1 = cluster1.sum / cluster1.length
    center2 = cluster2.sum / cluster2.length
  }
  // Display clusters
  println("Cluster 1:")
  println(data.toArray.filter(x =>
    math.abs(x - center1) < math.abs(x - center2)
  ).mkString(", "))

  println("Cluster 2:")
  println(data.toArray.filter(x =>
    math.abs(x - center2) <= math.abs(x - center1)
  ).mkString(", "))

  println("Center 1 = " + center1)
  println("Center 2 = " + center2)
  println("Anshu S035")
}
```

Output:

```
Cluster 1:
1.0, 2.0, 3.0
Cluster 2:
10.0, 11.0, 12.0
Center 1 = 2.0
Center 2 = 11.0
Anshu S035

Process finished with exit code 0
```